Directional occupancy and gain ladders across networks and batches

This snapshot contains the original MLP anchor and 11 completed continuations from the declared four-architecture, four-batch sweep. Incomplete jobs are excluded from numerical conclusions and remain recorded in the campaign status. Different parent checkpoints are explicitly labeled. Each continuation uses IID subsets and4096 updates. Plots are organized by architecture and batch, with no separate seed panels.

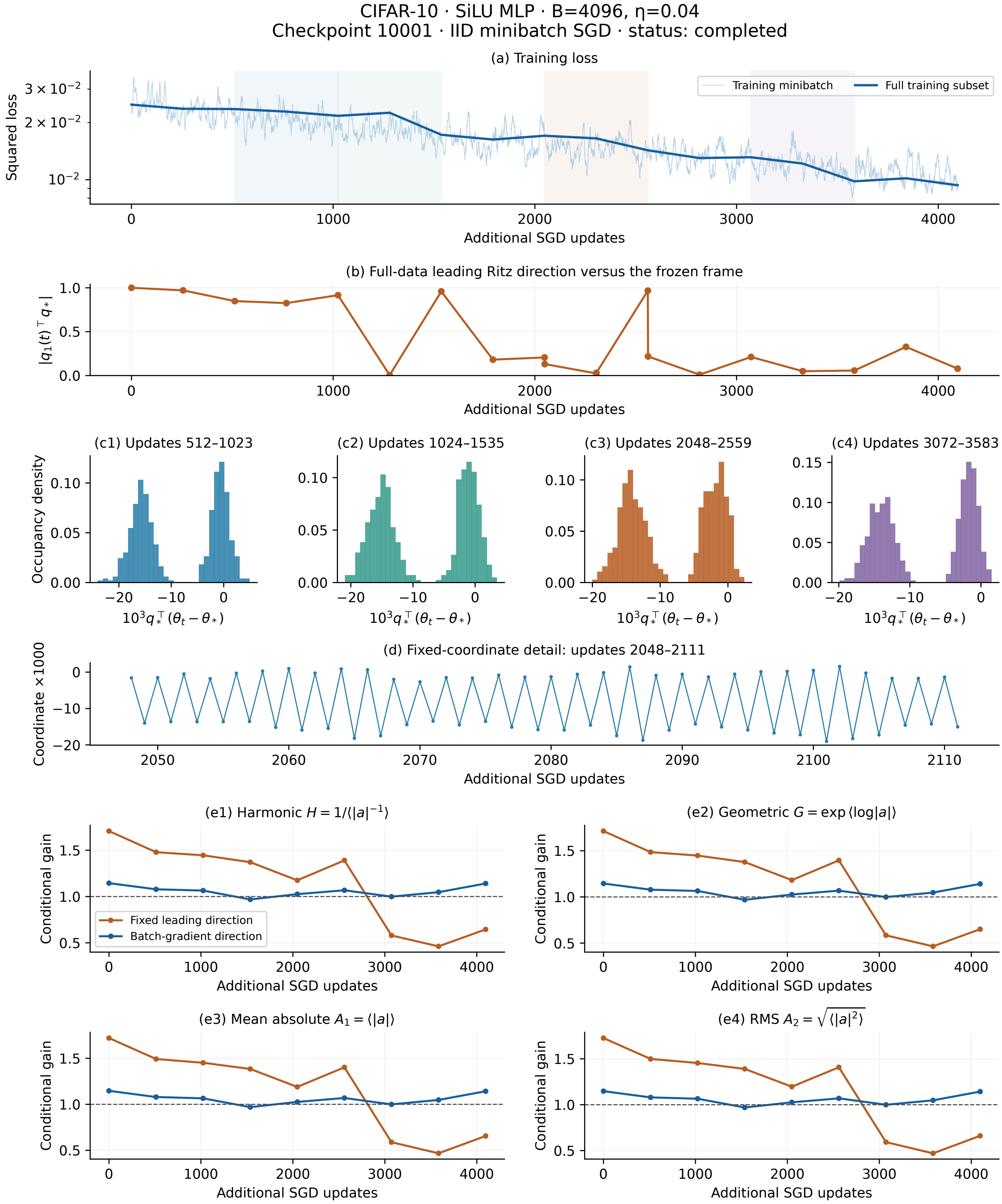

Figure 1: **SiLU MLP, batch 4096, learning rate 0.04, parent checkpoint 10001.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

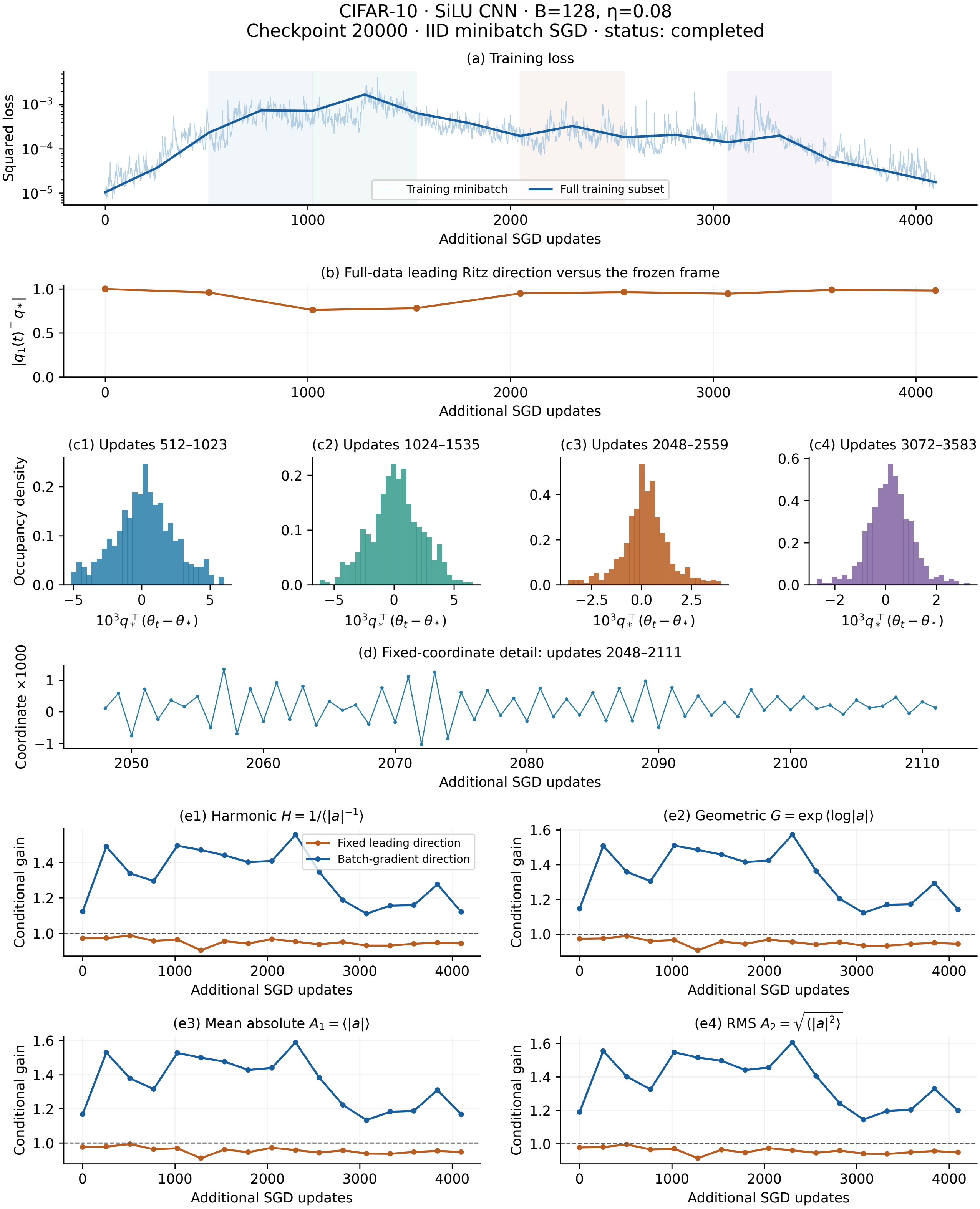

Figure 2: **SiLU CNN, batch 128, learning rate 0.08, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle |a|^{-1} \rangle$, $G = \exp\langle \log|a| \rangle$, $A_1 = \langle |a| \rangle$, and $A_2 = \sqrt{\langle |a|^2 \rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

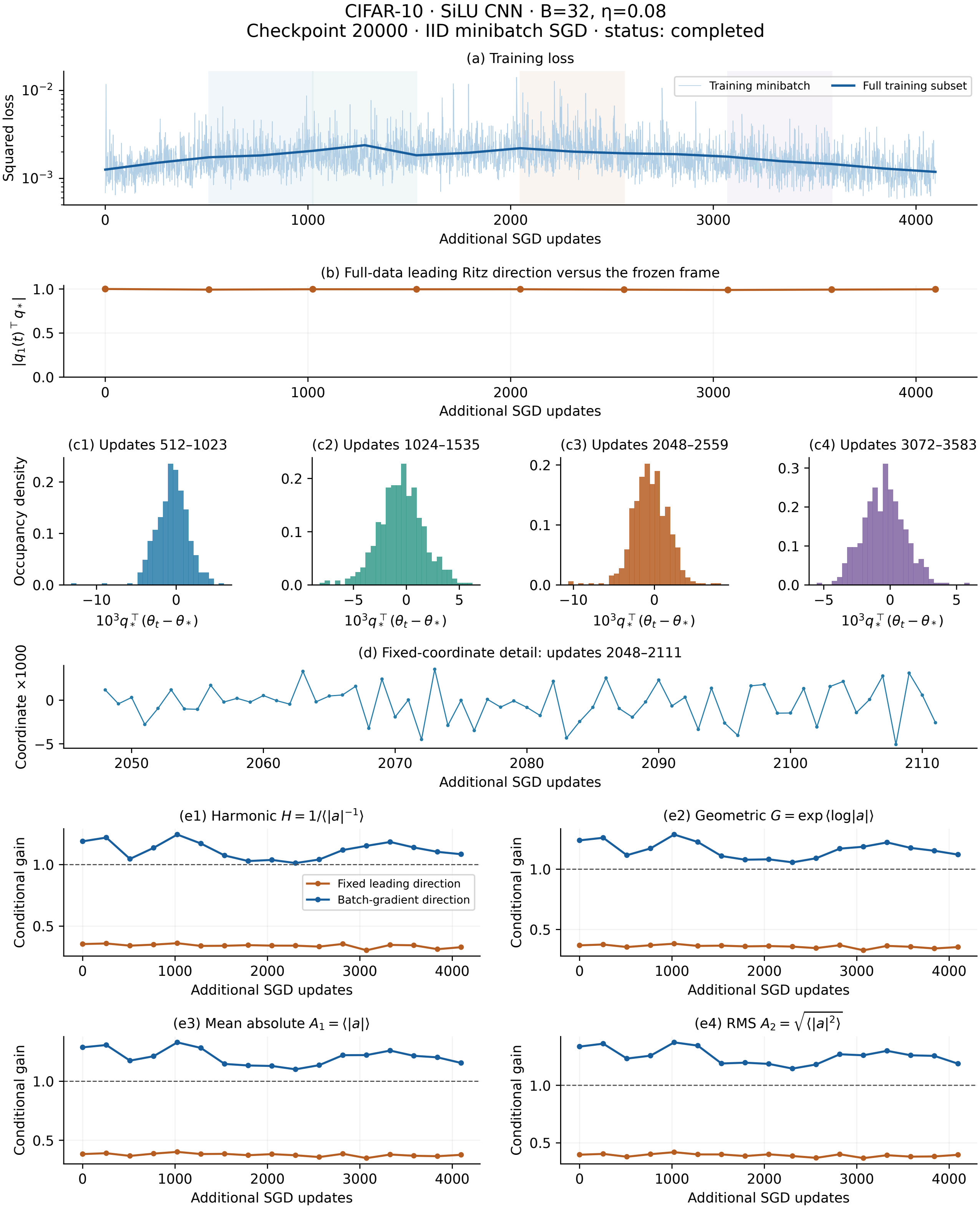

Figure 3: **SiLU CNN, batch 32, learning rate 0.08, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle |a|^{-1} \rangle$, $G = \exp\langle \log |a| \rangle$, $A_1 = \langle |a| \rangle$, and $A_2 = \sqrt{\langle |a|^2 \rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

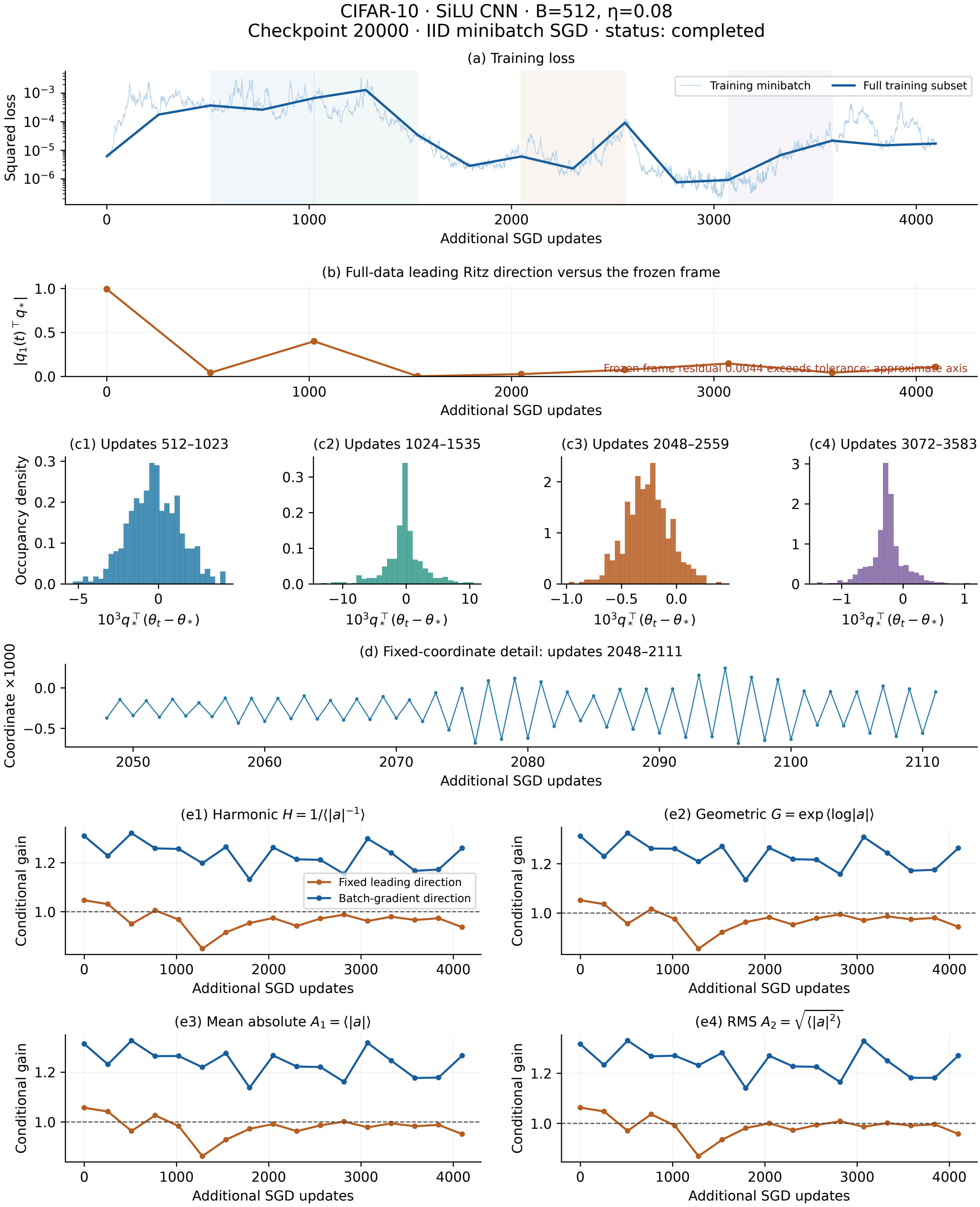

Figure 4: **SiLU CNN, batch 512, learning rate 0.08, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

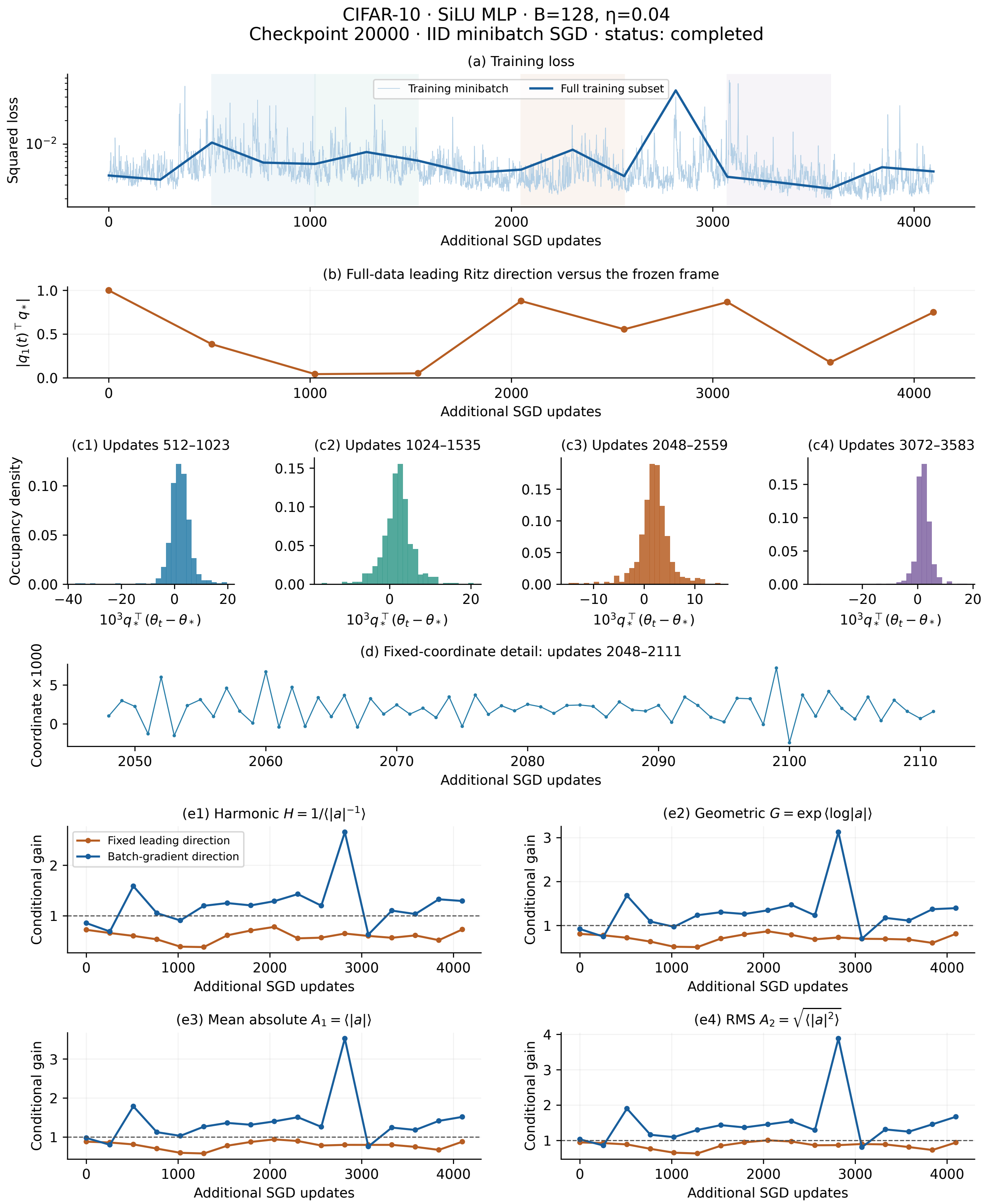

Figure 5: **SiLU MLP, batch 128, learning rate 0.04, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle |a|^{-1} \rangle$, $G = \exp\langle \log|a| \rangle$, $A_1 = \langle |a| \rangle$, and $A_2 = \sqrt{\langle |a|^2 \rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

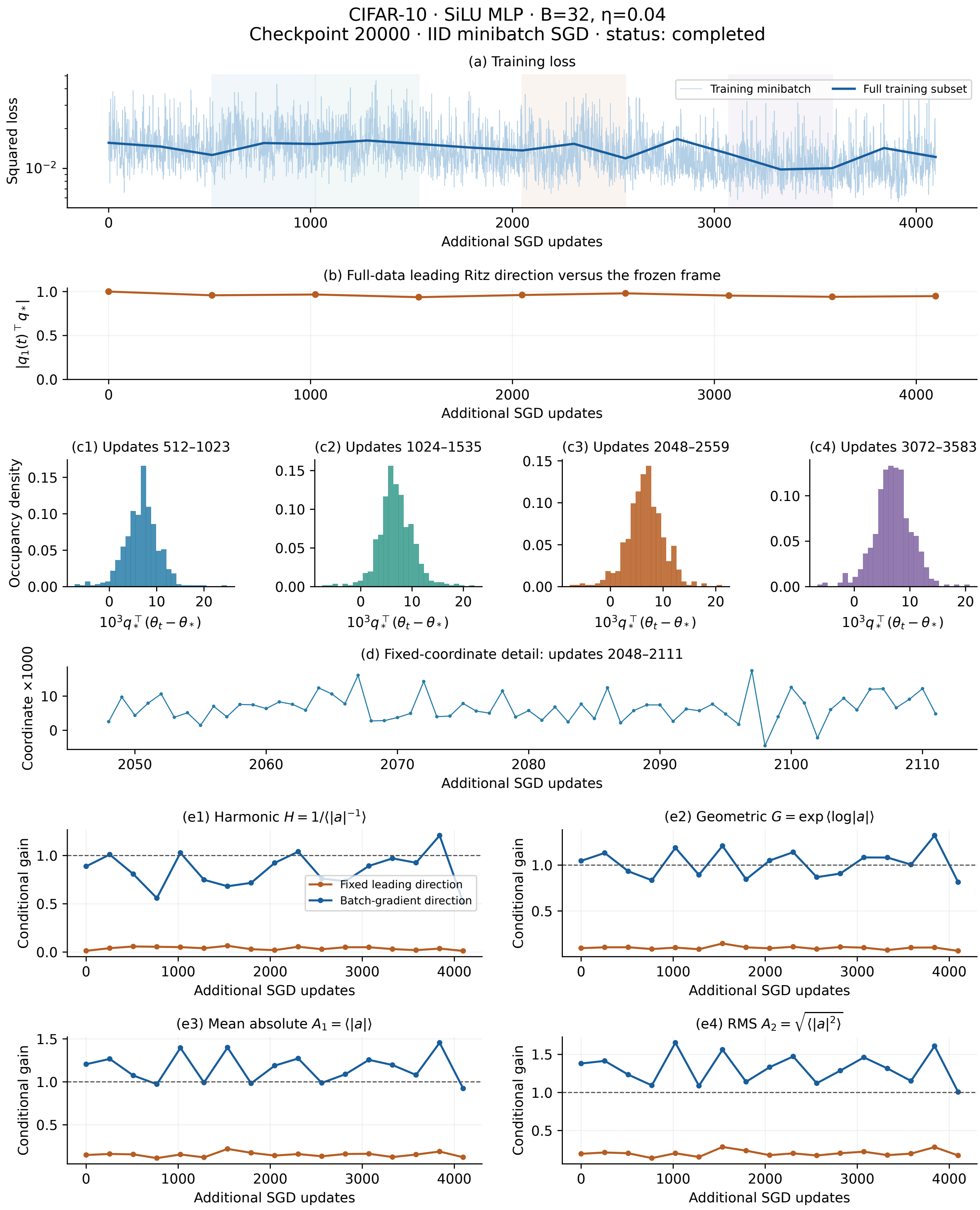

Figure 6: **SiLU MLP, batch 32, learning rate 0.04, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

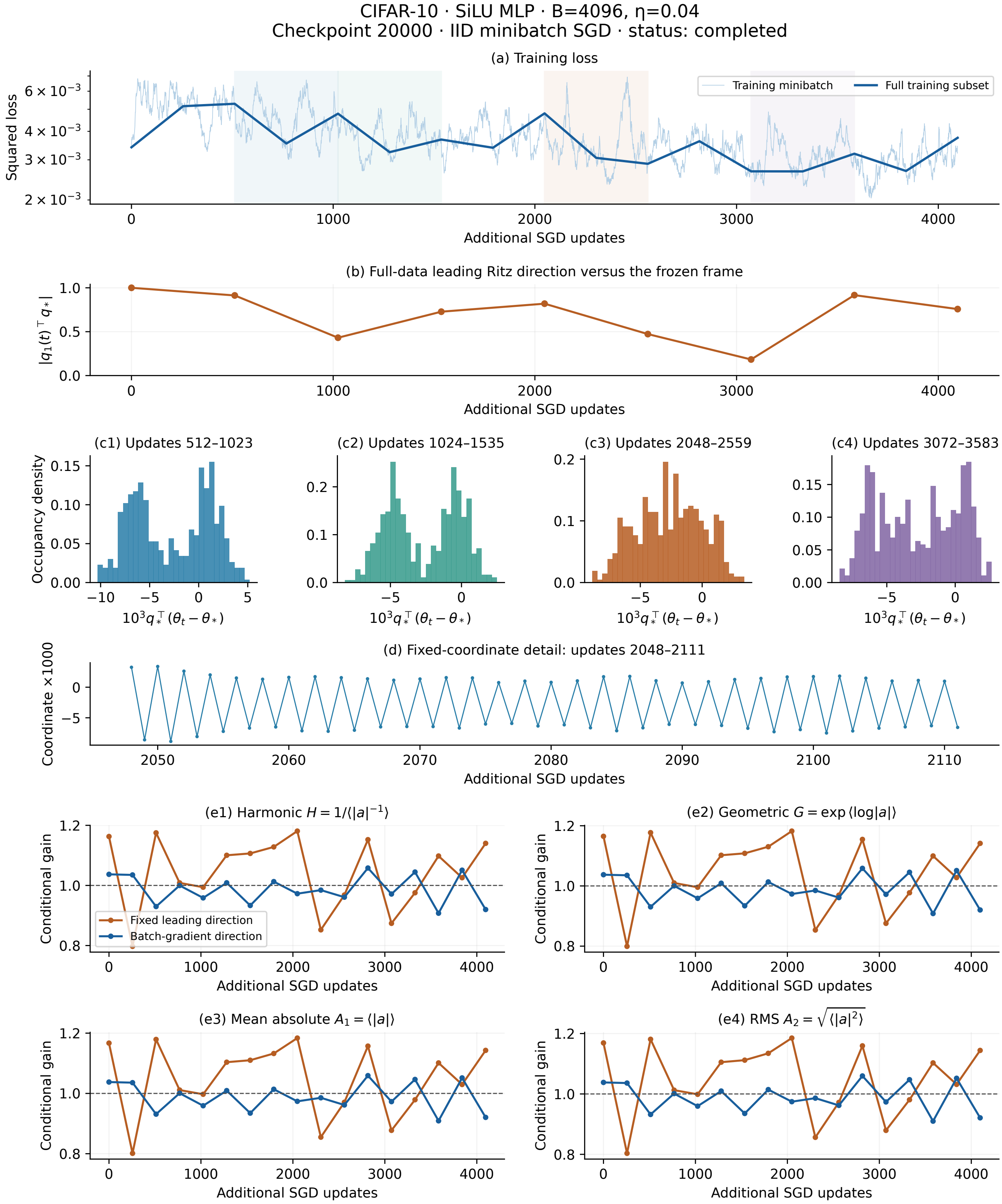

Figure 7: **SiLU MLP, batch 4096, learning rate 0.04, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

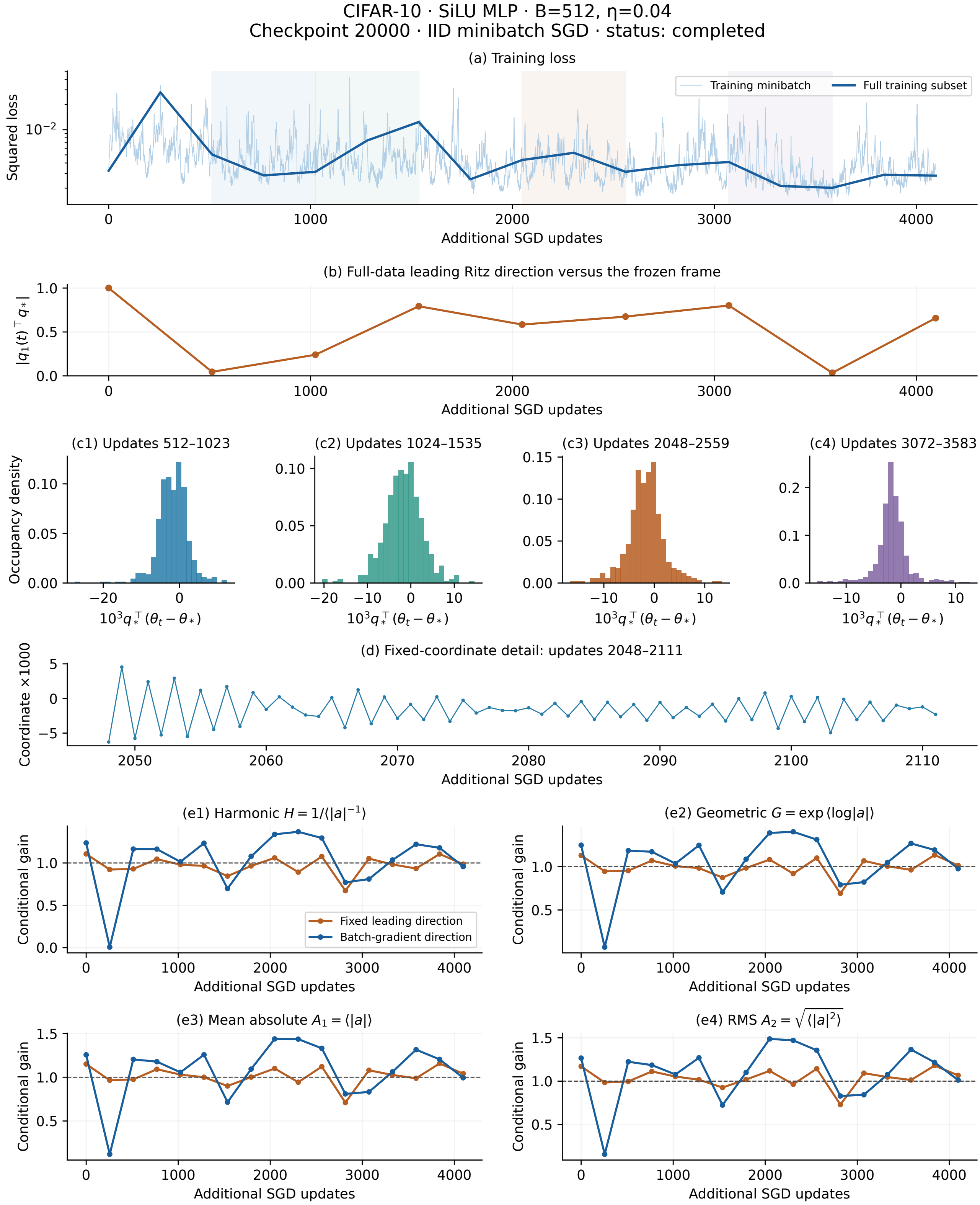

Figure 8: **SiLU MLP, batch 512, learning rate 0.04, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

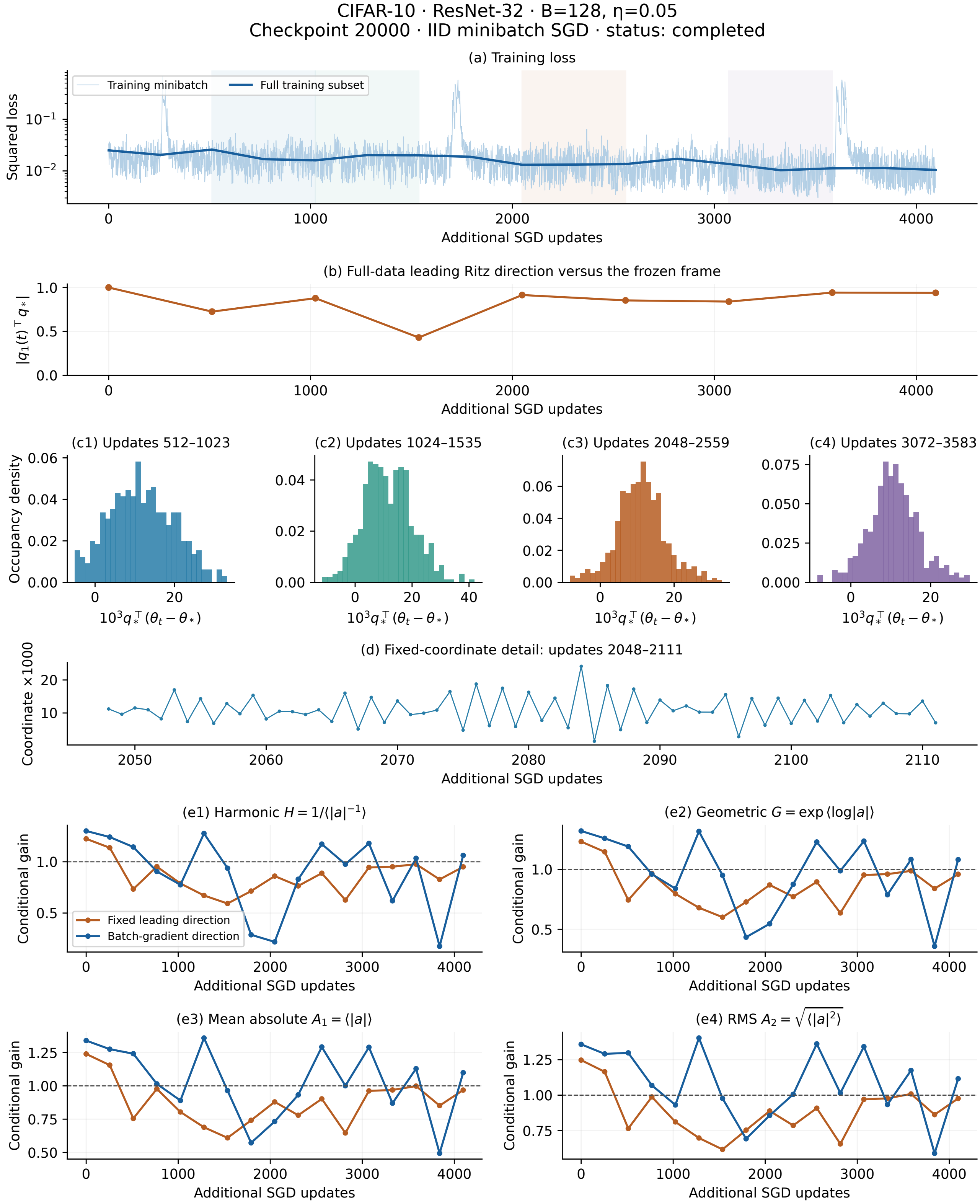

Figure 9: **ResNet-32, batch 128, learning rate 0.05, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

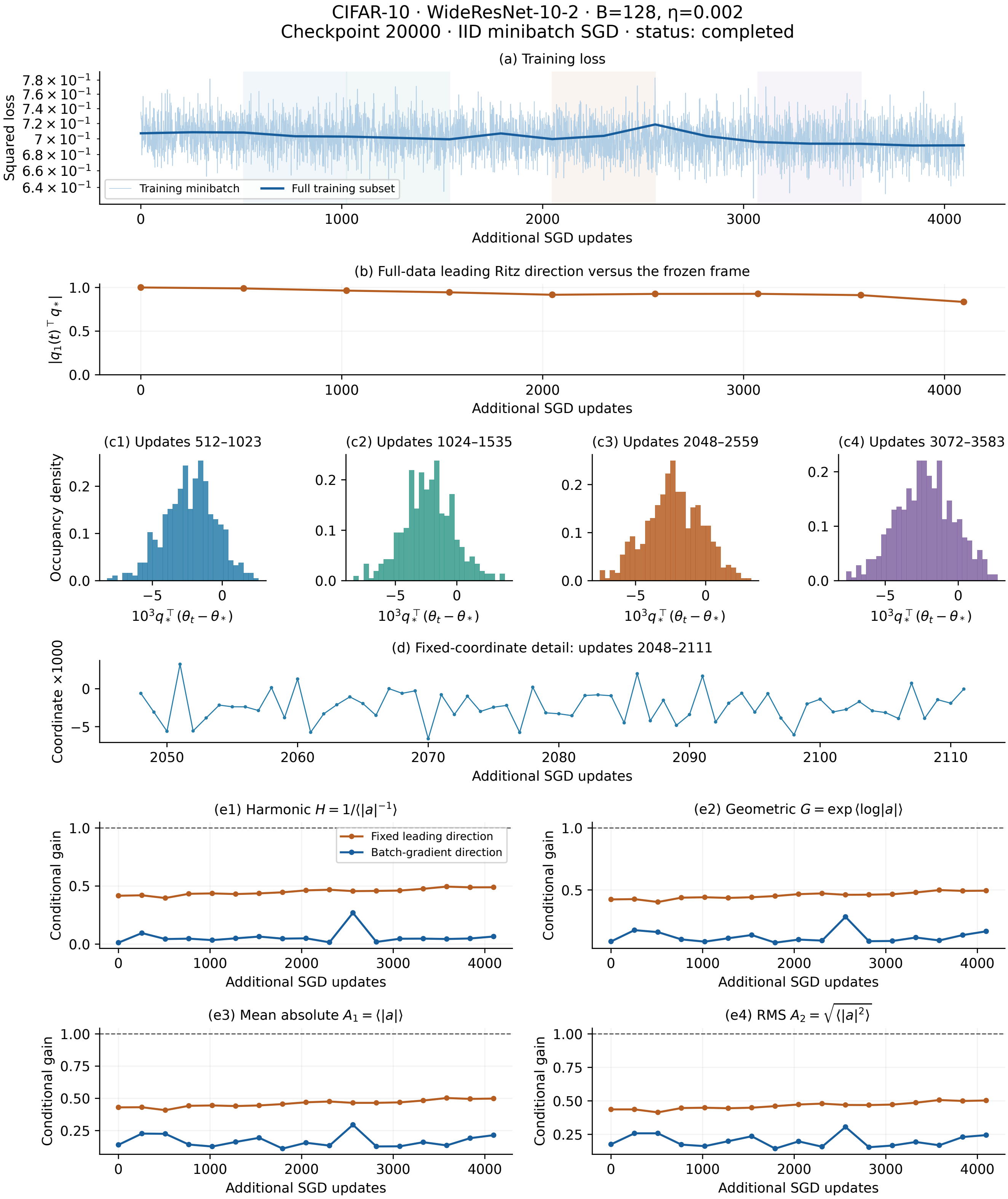

Figure 10: **WideResNet-10-2, batch 128, learning rate 0.002, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle |a|^{-1}\rangle$, $G = \exp\langle \log|a|\rangle$, $A_1 = \langle |a|\rangle$, and $A_2 = \sqrt{\langle |a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

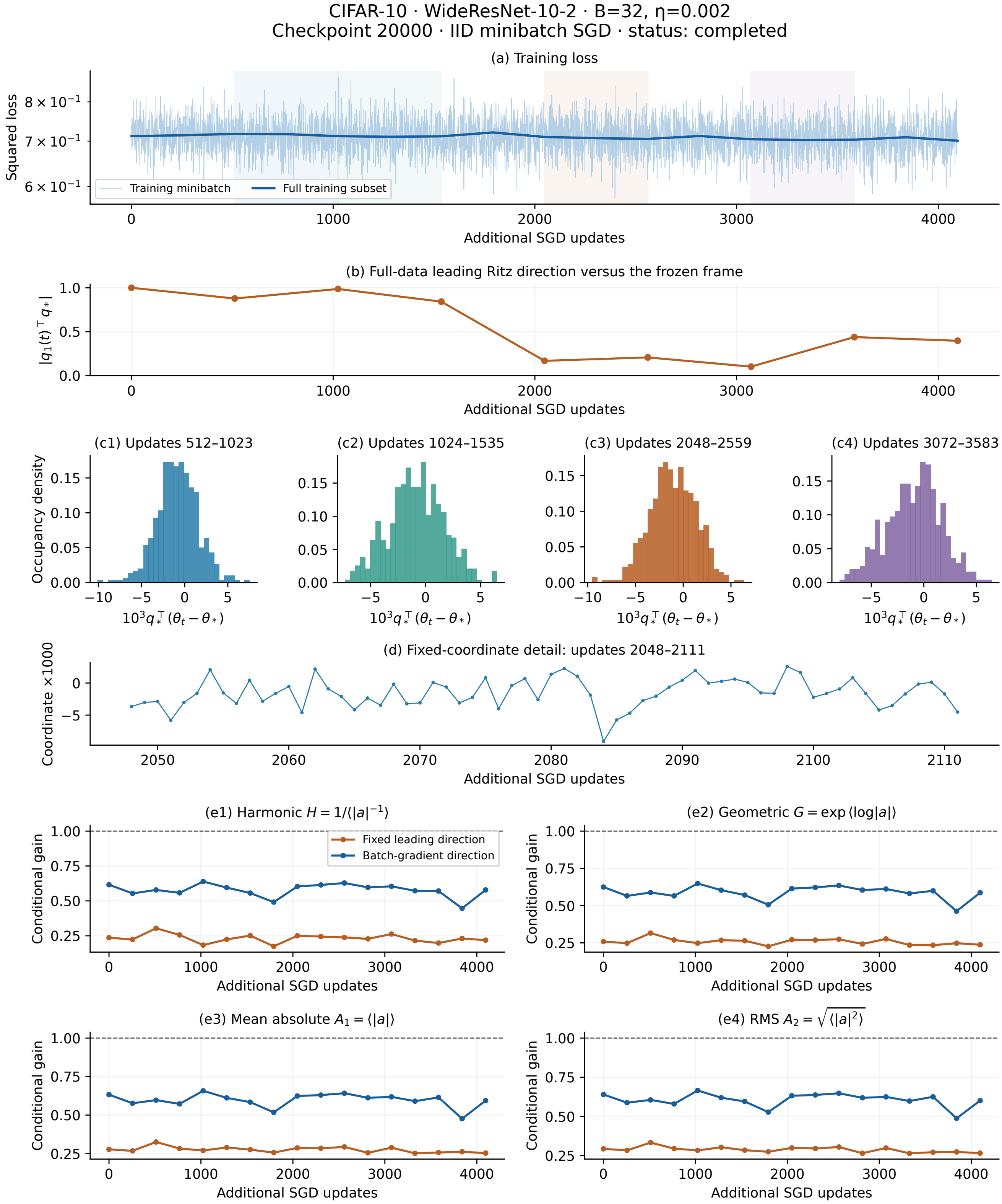

Figure 11: **WideResNet-10-2, batch 32, learning rate 0.002, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.

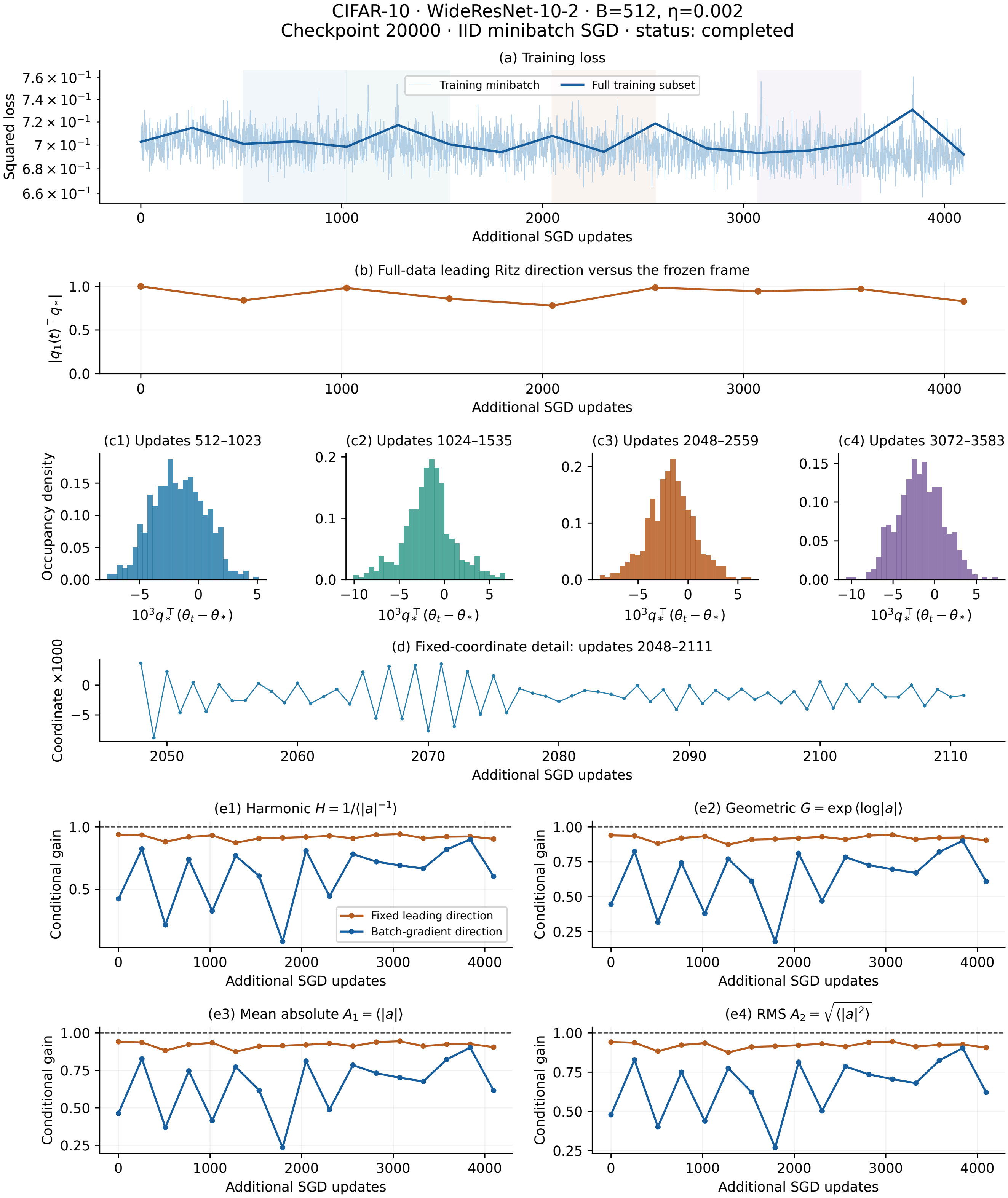

Figure 12: **WideResNet-10-2, batch 512, learning rate 0.002, parent checkpoint 20000.** Loss is shown for both training minibatches and the full 8192-example training subset. The plotting direction q_* and center θ_* stay fixed at the parent checkpoint; θ_* is not a verified minimizer. Alignment with separately recomputed leading Hessian Ritz vectors is measured on the full subset, not assumed. Four declared 512-update windows and one 64-update detail are displayed without selection by modality. The lower four panels show fresh conditional scalar gains, comparing $a_{*,B} = 1 - \eta q_*^\top H_B q_*$ and $a_{g,B} = 1 - \eta g_B^\top H_B g_B/(g_B^\top g_B)$. Their summaries are $H = 1/\langle|a|^{-1}\rangle$, $G = \exp\langle\log|a|\rangle$, $A_1 = \langle|a|\rangle$, and $A_2 = \sqrt{\langle|a|^2\rangle}$, all with reference one. Except for the original anchor, each state has 64 fresh probes every256 updates; the anchor uses 128 probes initially and24 every512 updates. Only even-update states appear here; no phase-averaged or stationary claim follows. No inverse-gain floor is applied.